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Variation in Repeated Handgrip Strength Testing Indicates Submaximal Force Production in Patients With Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

Stoyan Popkirov

Department of Neurology and Center for Translational and Behavioral Neurosciences, University Medicine Essen, University of Duisburg-Essen, Essen, Germany

Correspondence: Stoyan Popkirov (stoyan.popkirov@uk-essen.de)

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ABSTRACT

Background: Changes in handgrip strength have recently been adapted as clinical biomarkers for myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) under the assumption of a disease-specific peripheral neuromuscular dysfunction. However, some have proposed that strength impairments in ME/CFS are better explained by alterations in higher-order motor control. In serial measurements, exertion can been assessed through analysis of variation, since maximal voluntary contractions exhibit lower coefficients of variation (CV) than submaximal contractions.

Methods: Serial handgrip strength measurements of 105 ME/CFS patients and 66 healthy controls from a previously published biomarker validation study are analyzed post hoc regarding their CV. CV is separately compared in a subsample of participant with normal indexes of fatigability.

Results: Compared to healthy controls, patients had significantly higher CV, largely over the conservative 15% cutoff associated with submaximal exertion. In the subsample of study participants, whose within-session fatigability was within normal bounds, CV was still significantly higher in female patients; the difference in male patients was not statistically significant ($p = 0.06$).

Conclusions: This analysis suggests that loss of grip strength is likely compounded by alterations in higher-order motor control, challenging its utility as a biomarker of peripheral dysfunction. Functional weakness is discussed within a framework that sees motor fatigue as a result of reduced implicit self-efficacy acquired in the context of chronic dyshomoeostasis and disability.

1 | Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a complex condition characterized by persistent mental and physical fatigue, exertional intolerance, cognitive difficulties, widespread pain, and orthostatic intolerance [1]. Viral infections are common precipitants of the illness, yet the immunological and biobehavioral mechanisms of the chronic condition remain unknown. Identifying valid biomarkers has been one of the research avenues taken to improve understanding, outcome measurement and prognostication.

As weakness and fatigability are central hallmarks of ME/CFS, handgrip strength has been proposed as a suitable biomarker [2–4]. In a validation study, 105 patients with ME/CFS achieved lower mean and maximum grip strength and exhibited a steeper decline in strength across a series of trials compared to 66 healthy volunteers, though not compared to 18 patients with

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cancer-related fatigue [2]. In a cross-sectional study on 272 patients with ME/CFS, patients had a significantly larger drop in strength after the first of three handgrips compared to healthy controls, multiple sclerosis patients, or participants with chronic fatigue without an ME/CFS diagnosis [4]. The biological underpinning of this reduction of strength and endurance is currently a matter of debate.

In a recent deep phenotyping study, impairments in sustained grip strength in ME/CFS were correlated with the tendency to avoid effort in a behavioral task, but not with arm muscle mass, histological markers of disuse-related muscle atrophy, or an electromyographic index of peripheral muscle fatigue [5]. The decline in handgrip strength was thus attributed to "altered effort preferences." The implication that deficits were explained by a mere *preference* for lower exertion rather than an *inability* to fully exert was widely criticized [6]. The conclusions from the behavioral task have also been criticized on methodological grounds and are far from definitive. Regardless, the lack of correlation of grip strength with various markers of neuromuscular function challenged the assumption of a peripheral pathology, irrespective of the results of the effort task.

In studies on strength exertion, a higher variation in repeated strength production has been used as a sign of submaximal contraction [7]. The physiological assumption is that maximal contraction is a lower-order task of complete motor unit recruitment, while submaximal force production requires proprioceptively guided grading of strength, which entails higher-order complexity and, with it, increased variation of output. Handgrip strength testing in patients with ME/CFS can be reanalyzed in this light to assess whether within-session variation is abnormally high and thus indicative of higher-order motor control disruptions.

2 | Methods

2.1 | Participants and Study Design

The analysis used published raw data from a study by Jäkel et al. [2]. In it, 61 women und 44 men with ME/CFS (diagnosis based on Canadian Consensus Criteria) were recruited from a specialized university hospital outpatient clinic. Median duration of disease was 4 years (range: 1–42), 88% were classified as post-infectious ME/CFS. As a control group, 66 age and sex-matched self-reportedly healthy volunteers were tested. A second clinical group with cancer-related fatigue included just 18 female patients and no male patients and was not included in this post hoc analysis. Over two sessions (1 h apart), participants were asked to grip a digital hand dynamometer with maximum force in a standardized procedure. Each session consisted of ten three-second trials separated by 5 s of rest.

2.2 | Variation and Fatigability

The coefficient of variation of successive grip strength measurements was calculated as CV = standard deviation/mean × 100% [8]. A cutoff value of 15% has previously been shown to differentiate submaximal from maximal contractions in serial handgrip testing with a specificity of 94% [8]. Submaximal contractions can be expected in the context of pain, kinesiophobia, low motor self-efficacy, and other factors [9].

Peripheral fatigue would also increase variance, so it has to be accounted for. In the original study, fatigability is quantified through a "fatigue ratio" by dividing the maximum strength within one set of 10 trials by the mean value across the session. As this calculation does not consider the order of the maximum value, it would theoretically assign a high fatigue ratio to someone who progressively improves (indeed, in the first session, 11% of patients reached maximum handgrip strength in the second half of trials). Thus, for the purpose of this analysis, the sum of the first two measurements divided by the sum of the last two measurements within one session is used as an index of fatigability (IF). To minimize the influence of short-term fatigability on within-session variation, only patients with an IF within 2 SD of the mean IF in healthy controls of the respective gender will be included in a second analysis.

2.3 | Statistical Analysis

CV was tested for normality of distribution within groups and sessions using the Shapiro–Wilk Test. Between-group comparisons used median values and two-tailed Mann–Whitney *U* tests if one or both groups were not distributed normally. To explore the relationship between IF and CV, Spearman correlation coefficients were calculated and compared between groups using a Fisher *z*-test. To assess the effect of IF on CV group differences, partial Spearman rank correlations adjusting for IF as a covariate were calculated. The threshold of significance was set at $p < 0.05$.

3 | Results

To assess the variation of strength within each session, CV across the ten trials was calculated for each study participant.

TABLE 1 | Variation of handgrip strength across 10 consecutive trials in female healthy controls and patients.

	HC, female	ME/CFS, female	Mann–Whitney *U*		
	***n* = 36**	***n* = 61**	***U***	***z***	***p***
CV1 (%)	5.87 (4.46, 8.58)	14.73 (10.95, 20.39)	1955	6.40	< 0.00001
CV2 (%)	6.63 (5.57, 7.87)	14.88 (9.91, 21.85)	1822	5.41	< 0.00001

Note: Values are presented as median (interquartile range).
Abbreviations: CV, coefficient of variation across the ten trials of each session (1 and 2), calculated as standard deviation/mean × 100%; HC, healthy controls; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome.

Shapiro–Wilk testing showed that CV showed a significant departure from normality for all groups, genders, and sessions except for the first session in healthy males. Thus, the Mann–Whitney *U* test was used for all group comparisons. ME/CFS patients had a higher relative variation of strength (CV) compared to healthy controls across genders and sessions (Tables 1 and 2).

In a second analysis, in order to reduce the influence of abnormal within-session fatigability as a source of variation, only participants within two standard deviations of the mean IF of healthy controls for the respective gender were included (Tables 3 and 4). CV was significantly higher in female patients compared to controls in both sessions. The respective group difference in male participants did not reach statistical significance.

CV and IF correlated with moderate strength in both healthy controls and the subsample of patients with IF within normal bounds (HC: $r = 0.49$; ME/CFS: $r = 0.35$; both $p < 0.001$). There was no statistical between-group difference between the correlations (z-score: 1.27; $p = 0.20$), suggesting that the normal levels of fatigability in the filtered patient sample are unlikely to disproportionately affect their CV.

To examine this further, the influence of IF on the relationship between group membership (HC or ME/CFS patients) was quantified as the proportional reduction in the magnitude of Spearman rank correlation before and after accounting for IF as a covariate. In females, the moderate to strong correlation was reduced in magnitude by only 13.4% and 7.8% for sessions 1 and 2, respectively, while for males the reduction was substantial (60.1% and 19.4%).

4 | Discussion

In this post hoc analysis of serial grip strength measurements in patients with ME/CFS, a significantly higher coefficient of variation was found for all sessions and genders. Furthermore, even when only individuals without abnormal within-session fatigability were included, CV remained significantly higher in female patients compared to controls; in male patients, the respective group difference was not statistically significant with p-values around 0.06. In this filtered subset, within-session fatigability was not a major determinant of between-group difference in CV in females.

Although the utility of CV to reliably detect volitional reductions of effort is contested, its discriminatory power regarding maximal vs. submaximal contraction is established across a range of studies [8]. In a large study using five consecutive measurements

TABLE 2 | Variation of handgrip strength across 10 consecutive trials in male healthy controls and patients.

	HC, male	ME/CFS, male	Mann–Whitney *U*		
	$n = 30$	$n = 44$	*U*	*z*	*p*
CV1 (%)	5.82 (4.31, 7.66)	11.43 (6.97, 17.11)	1020	3.96	0.00008
CV2 (%)	5.94 (4.16, 7.26)	11.29 (6.69, 17.74)	1054	4.34	0.00001

Note: Values are presented as median (interquartile range).
Abbreviations: CV, coefficient of variation across the ten trials of each session (1 and 2), calculated as standard deviation/mean × 100%; HC, healthy controls; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome.

TABLE 3 | Variation of handgrip strength across 10 consecutive trials in female healthy controls and ME/CFS patients without an abnormal index of fatigue.

	HC, female	ME/CFS, female	Mann–Whitney *U*		
	$n_1 = 31, n_2 = 31$	$n_1 = 31, n_2 = 31$	*U*	*z*	*p*
CV1 (%)	5.44 (4.28; 6.61)	10.95 (8.24; 12.69)	864	5.40	< 0.00001
CV2 (%)	6.61 (5.42; 7.49)	9.99 (7.03; 11.89)	729	3.50	0.00048

Note: Values are presented as median (interquartile range). n_1 and n_2 denote sample sizes of included individuals from sessions 1 and 2.
Abbreviations: CV, coefficient of variation across the ten trials of each session (1 and 2), calculated as standard deviation/mean × 100%; HC, healthy controls; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome.

TABLE 4 | Variation of handgrip strength across 10 consecutive trials in male healthy controls and ME/CFS patients without an abnormal index of fatigue.

	HC, male	ME/CFS, male	Mann–Whitney *U*		
	$n_1 = 27, n_2 = 28$	$n_1 = 26, n_2 = 22$	*U*	*z*	*p*
CV1 (%)	5.69 (4.22; 7.38)	7.50 (5.67; 8.96)	458	1.90	0.05811
CV2 (%)	5.60 (4.14; 6.60)	6.69 (5.46; 8.29)	404	1.88	0.06198

Note: Values are presented as median (interquartile range). n_1 and n_2 denote sample sizes of included individuals from sessions 1 and 2.
Abbreviations: CV, coefficient of variation across the ten trials of each session (1 and 2), calculated as standard deviation/mean × 100%; HC, healthy controls; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome.

in healthy volunteers (88% female), the area under the curve (AUC) of the receiver operating characteristic for CV can be calculated as 0.868, suggesting high discriminatory accuracy [8]. At the threshold of CV ≥ 15%, only 6% of maximal contractions were misclassified as submaximal.

To interpret the significance of the higher-than-normal variation in repeated measurements, the basic strength parameters reported in the original study by Jäkel and colleagues should be considered first (2021). The mean grip strength in patients with ME/CFS is much lower than in healthy controls, and the peak-to-average strength ratio within sessions is much higher, suggesting acute fatigability. Remarkably, average strength is also lower than values reported in patients with mitochondrial myopathy [10, 11] and (non-ocular) myasthenia gravis [12, 13], two neurological diseases with pathophysiologically determined weakness and established short-term muscle fatigability [14]. In myasthenia gravis, strength largely recovers from fatigue induced by repetitive contractions within a few minutes [15]; By contrast, in the original ME/CFS study, reductions in strength were evident even after 1 h [2]. Electrophysiological characterizations of muscle and nerve function have not been able to elucidate this degree of muscle fatigability [16]. Peripheral fatigue through sustained maximal contraction can be assessed by measuring muscle fiber conduction velocity and electrically induced force responses before and after exertion, and is, by those measures, actually lower in ME/CFS patients compared to healthy controls [17]. In fibromyalgia syndrome, a clinical entity with significant overlap with ME/CFS [18, 19], hand grip strength is similarly reduced, and patients exhibit a characteristic discrepancy between peak and sustained grip strength as well as higher test–retest discrepancy [20], in line with the above-mentioned findings in ME/CFS. Studies combining maximum self-initiated contraction testing with superimposed transcutaneous electrical stimulation have demonstrated that the reduction in strength in fibromyalgia is associated with incomplete motor unit recruitment [21, 22]. A categorical difference in motor unit recruitment (complete vs. partial) is thought to produce increased variation in submaximal voluntary contraction as seen in the current analysis [7]. A study of 12 male ME/CFS patients using superimposed transcutaneous electrical stimulation previously found no abnormalities in maximum voluntary contraction, but the small sample was preselected to be able to achieve > 95% voluntary activation and had no measurable or clinically evident strength deficit [23]. Another study found twitch interpolation was required to record maximum quadriceps contraction in 35% of the ME/CFS patients compared to 20% of sedentary controls and 13% of depressed controls (difference not statistically significant in small study sample) [24]. In the future, this methodology could be employed in ME/CFS patients with relevant loss of strength as seen in handgrip measurements.

In a deep phenotyping study of ME/CFS, Walitt et al. [5] revealed that the ability of ME/CFS patients to maintain grip force over time was markedly reduced compared to HC, which was not explained by changes in lean arm mass or slow-to-fast muscle fiber atrophy (Type 2: Type 1 muscle fiber median Feret diameter ratio). Reduced strength endurance was, however, significantly correlated with the rate of choosing a harder task in an effort-expenditure for rewards task, suggesting a possible role of self-paced motor activation [5]. In a recent study using a similar task to assess preference biases to exert physical effort for reward, Valton et al. [25] found that subjects with depression exhibit a similar lower "effort acceptance bias,", which the authors computationally relate to trait-like exertion bias, rather than altered sensitivity to reward or effort, as is often assumed for depression. Interestingly, this effect was also seen in subjects whose depression was in remission at the time of testing, suggesting an independent personality trait or a "scarring" effect of an antecedent depressive episode [25].

Lastly, experimental findings from exercise science suggest that mental fatigue reduces self-paced power output and strength endurance, while increasing perceived exertion [26, 27]. Since mental fatigue is a key feature of ME/CFS, this, too, might contribute to the alterations of motor control mentioned above. The subjective feeling of fatigue has been conceptualized as a response to a critically reduced perception of allostatic self-efficacy, meaning the organism's self-assessed capacity to counter dyshomoeostasis [28]. It is plausible that chronic dyshomoeostasis can facilitate a generalization of low allostatic self-efficacy into related implicit decision-making such as motor self-efficacy, which determines self-initiated strength exertion. Rather than changing explicit "effort preferences," the experience of weakness undermines the predicted motor control efficacy, which can in turn alter motor output beyond volitional control [29, 30]. This predictive processing model of fatigue would also accommodate for associative learning perspectives on fatigue, as the assigned weight of predictions would be amenable to a range of influences [31].

Clinical experience makes it abundantly clear that the weakness seen in patients with ME/CFS is not a matter of "preference," poor motivation or resignation. So, how can the observed inconsistencies in motor output be understood more accurately? Findings from functional neurological disorder, a condition with a similarly fraught history of being mistaken for malingering [32], can help understand abnormal motor behavior enacted through "voluntary pathways" but without a corresponding sense of agency. A key finding in functional neurological disorder is that motor symptoms are associated with aberrant brain activations in sense-of-agency networks, particularly in the temporoparietal junction [33]. Changes in the mismatch analysis of intended and actual motor behavior can underlie the experience of involuntariness during movements that are performed through "voluntary" motor pathways. In the above-mentioned study by Walitt et al., ME/CFS patients showed analogous changes in temporoparietal junction activation during repeated handgrip strength testing, which was absent in heathy volunteers [5]. In clinical examinations of leg strength, the so-called Hoover sign reveals disruptions of self-agency in functional disorders and has been described anecdotally in ME/CFS [34]. In electrophysiological studies using pre-cued stimulus–response tasks, both patients with functional weakness and those with chronic fatigue exhibit similar changes in potentials associated with movement preparation (contingent negative variation in functional weakness; readiness potentials in chronic fatigue) [35–37]. Importantly, these changes are not found in healthy controls who willfully feign the same level of weakness as patients with functional symptoms [35], again underscoring the involuntary nature of these processes. Thus, from a neuroimaging and electrophysiological perspective the limb weakness

observed in ME/CSF might be closely related to the weakness seen in functional neurological disorder. Importantly, this proposition is based on the considerations above and should not be extrapolated to other neurological ME/CFS symptoms such as dizziness and pain, much less to the entire syndrome.

Efforts were undertaken to account for the influence of fatigability on variation in this post hoc analysis. However, regardless of such statistical considerations, the current electrophysiology literature does not support a separate peripheral neuromuscular or muscular cause for this fatigability. Indeed, altered biases of self-initiated strength production stemming from reduced motor self-efficacy would be prone to dynamic and self-iterative reductions across trials, as each exertion could further reinforce both explicit and implicit (embodied) expectancy of weakness and strain. This framing could also contribute to our understanding of post-exertional malaise. In the original study, one of the stronger and gender-independent correlations (albeit not corrected for multiple comparisons) was of the peak-to-average strength ratio in the second session ("Fatigue Ratio2", indicative of reduced control efficacy in this reading of the data) with self-reported post-exertional malaise [2]. The framework also offers an explanation for the reported worsening of symptoms in some ME/CFS patients during physical rehabilitation [38]: Exercises that push beyond individual thresholds of pain or fatigue can confirm and potently reinforce implicit sensorimotor biases of low motor self-efficacy. This is especially likely to happen if expressions of pain and fatigue are disregarded due to a categorical misunderstanding of ME/CFS. Fittingly, in a systematic review using individual patient data ($n = 959$), low self-efficacy (≤ 17 points on the Self Efficacy Scale) was the only patient characteristic predicting nonresponse to cognitive behavioral therapy [39].

In conclusion, the increased variation in maximal self-initiated handgrip strength in ME/CFS patients suggests alterations in higher-order motor control as a cause of limb weakness in ME/CFS. Recent experimental studies suggest that these changes in force production are not enacted on a consciously accessible level of neural computation but emerge from sensorimotor decision-making related to reductions in implicit motor self-efficacy. Treatment should accordingly aim at iterative reinforcement of explicit and implicit self-efficacies of movement by demonstrating untapped strength reserves, addressing maladaptive exertion avoidance and providing guidance on coping and recovery strategies.

Author Contributions

Stoyan Popkirov: conceptualization, writing – original draft, methodology, writing – review and editing, formal analysis.

Ethics Statement

This is a post hoc analysis of data that was previously published Open Access under a Creative Commons Attribution 4.0 International License. The original study was approved by the ethics committee of the Charité (EA2/038/14 from 12th August 2015).

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Raw data from original study by Jäkel et al. [2] available at https://static-content.springer.com/esm/art%3A10.1186%2Fs12967-021-02774-w/MediaObjects/12967_2021_2774_MOESM6_ESM.xlsx. No additional datasets were generated during the current study.

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